



Yuyang Chen

AI for EDA | Computational Lithography | Algorithm R&D
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Profile

Ph.D. student at ShanghaiTech University, expected to graduate in 2027, with algorithm development and validation experience in semiconductor manufacturing and manufacturing EDA. Research spans layout understanding, mask optimization and GPU-accelerated lithography simulation. Long-term focus: connect these stages through shared data, physics-aware models and full-flow validation to improve manufacturability and computational efficiency, with agentic AI workflows as a future research direction.

Industry Experience

Leading manufacturing EDA company in China Jul 2026 - Sep 2026

Algorithm R&D Intern | AI models for inverse lithography (ILT)

- Implemented and validated AI models for inverse lithography, producing working code and evaluation results for the target task.

Major semiconductor foundry in China Jul 2025 - Sep 2026

Algorithm R&D Intern | Detection in semiconductor manufacturing

- Developed and validated algorithms for a detection project in semiconductor manufacturing, producing a working implementation and experimental results.

Selected Research Projects

GPU-Native Numerical-Neural 3D Lithography Simulation

N3Litho | DAC 2026 | First author, equal contribution

- Co-developed a GPU-native 3D lithography framework combining a vectorial Abbe solver with neural field reconstruction from image-plane fields and mask/film-stack parameters, supporting resist and OPC/ILT evaluation.
- **Validation:** In the paper's 22 nm setup, achieved $29.91 \mu\text{m}^2/\text{s}$ throughput versus 0.0186 for the commercial reference, with 3D aerial-image MSE of 6.6×10^{-4} .

Scaling Vision Models to Full-Chip Mask Optimization

LVM-MO | DAC 2025 | Co-first author | Best Paper Award Nomination

- Co-developed a full-chip mask optimization model using geometry pretraining, differentiable lithography feedback and cross-scale training on approximately 5M layouts to address long-range interactions and stitching errors.
- **Validation:** On the reported benchmark, optimized 60×60 and $150 \times 150 \mu\text{m}^2$ layouts in 4 s and 28 s on one A100 GPU, demonstrating large-layout inference beyond training scales.

Manufacturability-Aware Manhattan Mask Generation

TokMan | NeurIPS 2025 | Co-first author

- Co-developed a Manhattan mask optimization method based on geometric sequence generation, combining a Diffusion Transformer, differentiable lithography feedback and lightweight ILT refinement.
- **Validation:** On ICCAD13-S, reduced EPE from 2.70 to 2.16 and average mask shot count from 4948.4 to 287.6 versus IL-ILT, evaluating print fidelity, mask-writing complexity and runtime together.

Selected Research Projects Continued

Structured Layout Understanding and Reliable Hotspot Detection

LLM-HD, DAC 2024, first author (equal contribution) | HyDAS, IEEE TCAD, first author

- Developed hotspot-detection methods: LLM-HD preserves geometry through GDS semantic encoding; HyDAS combines spatial/spectral features, deformed attention and selective prediction to support engineer review of high-risk patterns.
- **Validation:** LLM-HD raised mean hotspot recall from 91.99% to 93.99% versus DATE'22 on the reported benchmarks and received a DAC 2024 Best Paper Award Nomination. HyDAS includes the public ICCAD2020 dataset.

Multimodal Pretraining for Label-Efficient Hotspot Detection

From Contrastive to Generative Alignment | IEEE TCAD 2026 | Second author

- Contributed to multimodal pretraining research aligning layout images and GDSII text through contrastive learning and polygon-coordinate generation. The project uses 500K paired samples and tests low-resource adaptation and cross-design generalization to address labeling cost and model reuse.

Evaluating AI Across the Computational Lithography Flow

Ariadne's Thread | ICCAD 2026, forthcoming | Co-author

- Contributed to full-flow benchmark research connecting lithography modeling, OPC, EBL modeling and MPC, with aligned intermediate data and commercial-tool re-validation of composed pipelines.
- **Results:** Over 104K aligned instances across 28/32/45 nm nodes and 18 method combinations. In 4 of 5 tested conditions, the best full-flow combination differed from the combination of stage-wise winners, motivating cross-stage optimization.

Earlier Research Experience

Research Assistant, ShanghaiTech University

2020 - 2022

Supervisor: Pingqiang Zhou. Investigated 3D-IC thermal stress, including TSV/strained-silicon interactions, mobility effects and keep-out-zone implications. Second author, *Integration* 88 (2023), 196–202; published online in 2022.

Education and Coursework

ShanghaiTech University | Integrated master's-to-Ph.D. track

2022 - 2027 (expected)

Electronic Science and Technology; doctoral stage since 2025. Advisor: Prof. Hao Geng.

ShanghaiTech University | B.Eng., Electronic Information Engineering

2016 - 2022

Graduate coursework: Deep Learning; Electronic Design Automation; AI4IC; Digital Image Processing; Introduction to Quantum Information; Intelligent Computing Systems.

Undergraduate coursework: Linear Algebra; Advanced Mathematics; Discrete Mathematics; Computer Programming; Circuit Fundamentals; Electromagnetics; Signals and Systems; Analog Circuits; Digital Circuits; Semiconductor Device Fundamentals.

Skills and Additional Recognition

Methods and tools: Python, PyTorch, PyTorch Geometric, Linux, L^AT_EX, AI-assisted development (Vibe Coding); GDS representations, differentiable simulation, GPU acceleration and model evaluation.

Patent: Co-inventor, hotspot-detection model training and detection method; published Chinese patent application CN 119494313 A (2025).

Training and service: Cadence Digital and Signoff Full Flow Tools Training (Jul 2024); Outstanding Teaching Assistant, SIST, ShanghaiTech University (2024).